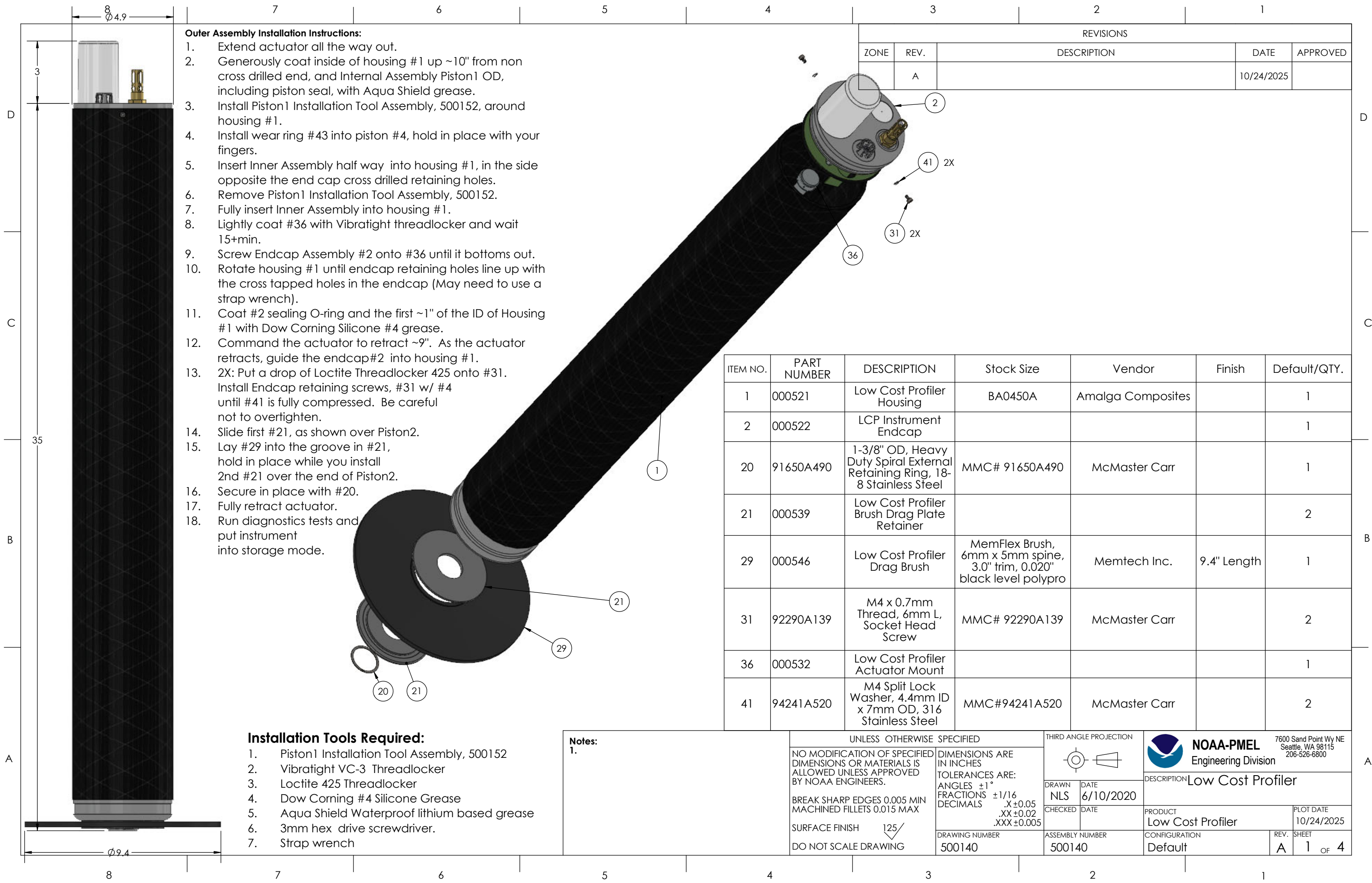




Outer Assembly Installation Instructions:

1. Extend actuator all the way out.
2. Generously coat inside of housing #1 up ~10" from non cross drilled end, and Internal Assembly Piston1 OD, including piston seal, with Aqua Shield grease.
3. Install Piston1 Installation Tool Assembly, 500152, around housing #1.
4. Install wear ring #43 into piston #4, hold in place with your fingers.
5. Insert Inner Assembly half way into housing #1, in the side opposite the end cap cross drilled retaining holes.
6. Remove Piston1 Installation Tool Assembly, 500152.
7. Fully insert Inner Assembly into housing #1.
8. Lightly coat #36 with Vibratight threadlocker and wait 15+min.
9. Screw Endcap Assembly #2 onto #36 until it bottoms out.
10. Rotate housing #1 until endcap retaining holes line up with the cross tapped holes in the endcap (May need to use a strap wrench).
11. Coat #2 sealing O-ring and the first ~1" of the ID of Housing #1 with Dow Corning Silicone #4 grease.
12. Command the actuator to retract ~9". As the actuator retracts, guide the endcap#2 into housing #1.
13. 2X: Put a drop of Loctite Threadlocker 425 onto #31. Install Endcap retaining screws, #31 w/ #4 until #41 is fully compressed. Be careful not to overtighten.
14. Slide first #21, as shown over Piston2.
15. Lay #29 into the groove in #21, hold in place while you install 2nd #21 over the end of Piston2.
16. Secure in place with #20.
17. Fully retract actuator.
18. Run diagnostics tests and put instrument into storage mode.

Installation Tools Required:

1. Piston1 Installation Tool Assembly, 500152
2. Vibratight VC-3 Threadlocker
3. Loctite 425 Threadlocker
4. Dow Corning #4 Silicone Grease
5. Aqua Shield Waterproof lithium based grease
6. 3mm hex drive screwdriver.
7. Strap wrench

Notes:

1.

REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A		10/24/2025	

ITEM NO.	PART NUMBER	DESCRIPTION	Stock Size	Vendor	Finish	Default/QTY.
1	000521	Low Cost Profiler Housing	BA0450A	Amalga Composites		1
2	000522	LCP Instrument Endcap				1
20	91650A490	1-3/8" OD, Heavy Duty Spiral External Retaining Ring, 18-8 Stainless Steel	MMC# 91650A490	McMaster Carr		1
21	000539	Low Cost Profiler Brush Drag Plate Retainer				2
29	000546	Low Cost Profiler Drag Brush	MemFlex Brush, 6mm x 5mm spine, 3.0" trim, 0.020" black level polypro	Memtech Inc.	9.4" Length	1
31	92290A139	M4 x 0.7mm Thread, 6mm L, Socket Head Screw	MMC# 92290A139	McMaster Carr		2
36	000532	Low Cost Profiler Actuator Mount				1
41	94241A520	M4 Split Lock Washer, 4.4mm ID x 7mm OD, 316 Stainless Steel	MMC#94241A520	McMaster Carr		2

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BREAK SHARP EDGES 0.005 MIN
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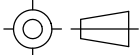
SURFACE FINISH 125

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ANGLES ±1°
FRACTIONS ±1/16
DECIMALS .X±0.05
.XX±0.02
.XXX±0.005

DRAWING NUMBER
500140

THIRD ANGLE PROJECTION



DRAWN NLS
DATE 6/10/2020

CHECKED DATE

ASSEMBLY NUMBER
500140



NOAA-PMEL
Engineering Division

7600 Sand Point Wy NE
Seattle, WA 98115
206-526-6800

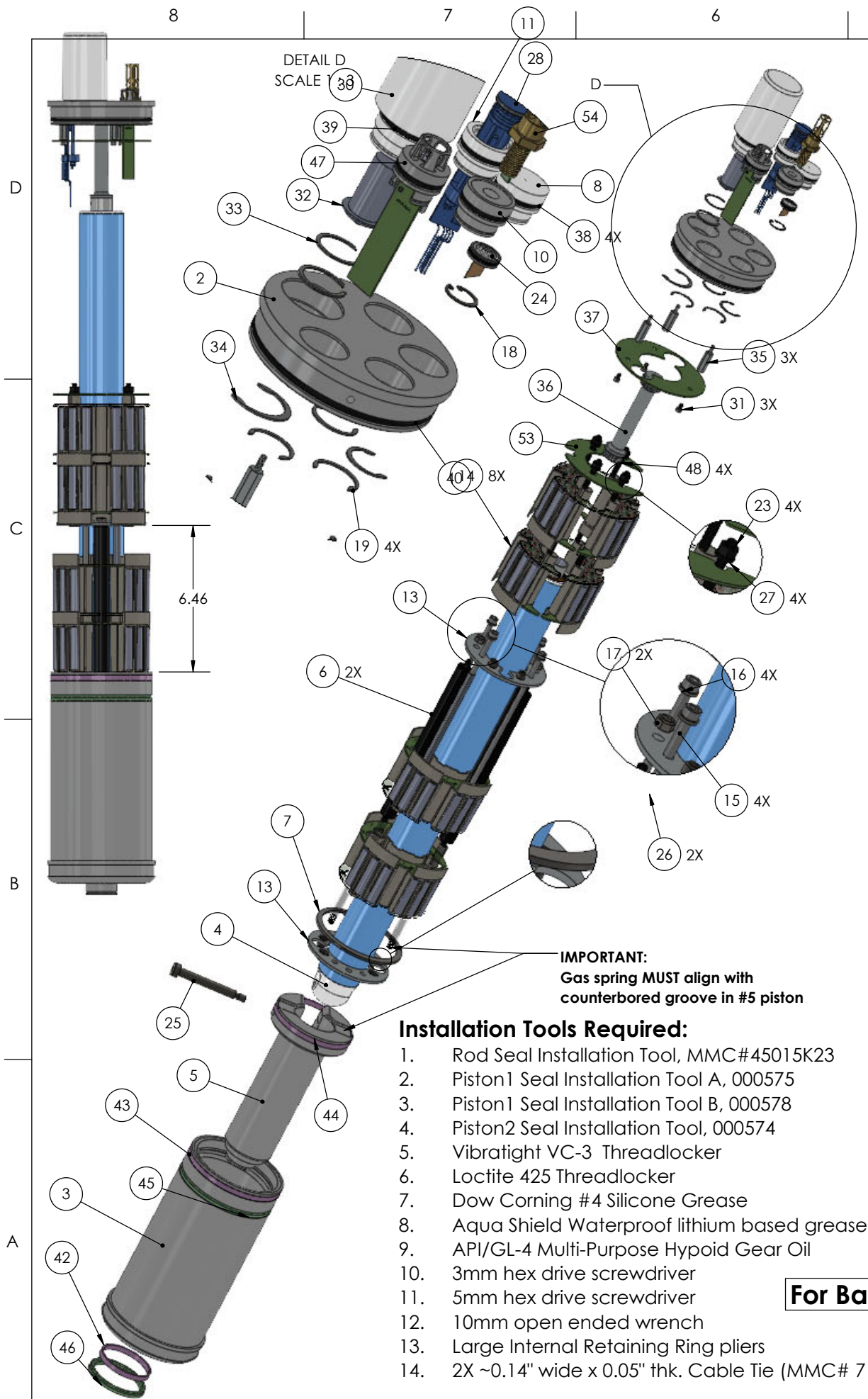
DESCRIPTION
Low Cost Profiler

PRODUCT
Low Cost Profiler

CONFIGURATION
Default

REV. SHEET
A 1 OF 4

PLOT DATE
10/24/2025



ITEM NO.	PART NUMBER	DESCRIPTION	Stock Size	Vendor	Finish	Internal Assembly/QTY.
2	000522	LCP Instrument Endcap				1
3	000524	Low Cost Profiler Piston #1	PVC Type I Rod			1
4	FA-BH50-XXX-12-12	Firgelli Bullet Series 50 Cal. Linear Actuator, 12" stroke	FA-BH50-XXXX-12-12	Firgelli Automation		1
5	000525	Low Cost Profiler Piston #2				1
6	89U140341TT0667	IGS 8/18 x 140mm Gas Spring	89U140341TT0667	Industrial Gas Spring		2
7	98455A161	100mm Steel Internal Retaining Ring	MMC# 98455A161	McMasterCarr		1
8	000531	9LX Pressure Port	Type I			1
9	000531	RBR Coda Sensor Port	Type I			1
10	000531	7/16-20 Threaded Port	Type I			1
11	000531	Turner Cyclops CHLA Sensor Port	Type I			1
12	RBRcoda_T	RBR Coda T.	OEM RBRcoda T	RBR Global		1
13	500142	Low Cost Profiler Gas Spring Ring				2
14	500147	Low Cost Profiler 9 cell AA Battery Pack				8
15	91290A155	M6 x 1mm , 200mm L, Alloy Steel Socket Head Screw	M6 x 1mm x 200m L SHCS, MMC# 91290A155	McMaster Carr		4
16	91166A250	M6 , 6.4 mm ID, 12 mm OD, Zinc-Plated Steel Washer	MMC# 91166A250	McMaster Carr		4
17	90576A115	M6 x 1mm, Medium-Strength Steel Nylon-Insert Locknut	90576A115	McMaster Carr		2
18	98455A224	19mm ID Internal Retaining Ring, Black-Phosphate Spring Steel	MMC# 98455A224	McMaster Carr		1
19	97414A705	1" OD Low-Clearance External Retaining Ring, Spring Steel	MMC# 97414A705	McMaster Carr		4
23	93369A141	M6 x 1mm, Nylon Hex Nut	MMC# 93800A600	McMaster Carr		4
24	Keller 9LX Pressure Sensor	Keller 9LX- 30 Bar OEM Pressure Sensor	9LX-30 Bar	Keller America		1
25	92981A309_ALLOY STEEL SHOULDER SCREW	M8 x 1.25mm, 10mm Dia x 70mm L, Shoulder Screw	92981A309	McMaster Carr		1
26	000545	Low Cost Profiler Gas Spring Spacer	5/16" OD x 0.016" wall, MMC#7237K33	McMaster Carr	7.200" Long	2
27	95610A370_NYLON PLASTIC WASHER_	M6, 6.4mm ID, x 12mm OD, Nylon Plastic Washer	MMC# 95610A570	McMaster Carr		4
28	2110-000-C-NH	Turner Cyclops 7F CHLA, No Housing OEM	Cyclops 7F- No Housing Version	Turner Designs Inc.		1
30	000547	Low Cost Profiler Antenna Housing	Type I			1
31	92290A139	M4 x 0.7mm Thread, 6mm L, 316 Stainless Steel Socket Head Screw	MMC# 92290A139	McMaster Carr		3
32	HC600E	Tallysman OEM Iridium Helix Antenna	HC600E	Tallysman Wireless		1
33	92131A630_SINGLETURN SPIRAL INTERNAL RETAINING RING	28mm ID, Single-Turn Spiral Internal Retaining Ring, 1060-1090 Spring Steel	MMC# 92131A630	McMaster Carr		2
34	98555A230	1-3/8" OD Side-Mount External Retaining Ring, Steel	MMC# 98555A230	McMaster Carr		1
35	98952A158	M4 x 0.70mm x25mm L w/ 8mm Hex, Male-Female Threaded Hex Standoff, Aluminum	MMC# 98952A158	McMaster Carr		3
36	000532	Low Cost Profiler Actuator Mount				1
37	Low Cost Profiler Main Control Board	Low Cost Profiler Main Control Board				1
38	9452K84	2-120 Buna N Oring	2-120, MMC# 9452K84	McMaster Carr		4
39	9452K89	2-125 Buna N Oring	2-125, MMC# 9452K89	McMaster Carr		1
48	000542	Low Cost Profiler Battery Rod	M6 x 1.0mm Nylon 6/6 Threaded Rod, MMC# 98882A249	McMaster Carr	6.25" Long	4
40	9452K194	2-244 Buna N Oring	2-244, MMC# 9452K194	McMaster Carr		1
42	0307-PDT-H-093-0250	7.31" Long	0307-PDT-H-093-0250	Parker	7.31" Long	1
43	0307-PDT-H-093-0250	13.81" Long	0307-PDT-H-093-0250	Parker	13.81" Long	1
44	0307-PDT-H-093-0250	11.44" Long	0307-PDT-H-093-0250	Parker	11.44" Long	1
45	Parker_N4274_250_04000	4" ID Parker PolyPak SPP Hydraulic Seal	N4274-250-04000	Parker		1
46	Parker_N4274_250_02250	2.25" ID Parker PolyPak SPP Hydraulic Seal	N4274-250-02250	Parker		1
47	500126	LFS1107 Conductivity Sensor				1
53	Low Cost Profiler Battery Power Management Board	Low Cost Profiler Batery Power Management Board				1
54	SoundNine_Digital_OEM_Temperature_Sensor	Sound Nine Digital OEM Temperature Sensor	OEM Temp	Sound Nine		1

For Battery Pack Option See Sheet #3

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BREAK SHARP EDGES 0.005 MIN MACHINED FILLETS 0.015 MAX

SURFACE FINISH 125

DO NOT SCALE DRAWING

DIMENSIONS ARE IN INCHES

TOLERANCES ARE:

ANGLES ±1°

FRACTIONS ±1/16

DECIMALS .X±0.05

.XX±0.02

.XXX±0.005

DRAWING NUMBER 500140

ASSEMBLY NUMBER 500140

THIRD ANGLE PROJECTION

DRAWN NLS

DATE 6/10/2020

CHECKED

DATE

NOAA-PMEL Engineering Division

7600 Sand Point Wy NE Seattle, WA 98115 206-526-6800

DESCRIPTION

Low Cost Profiler

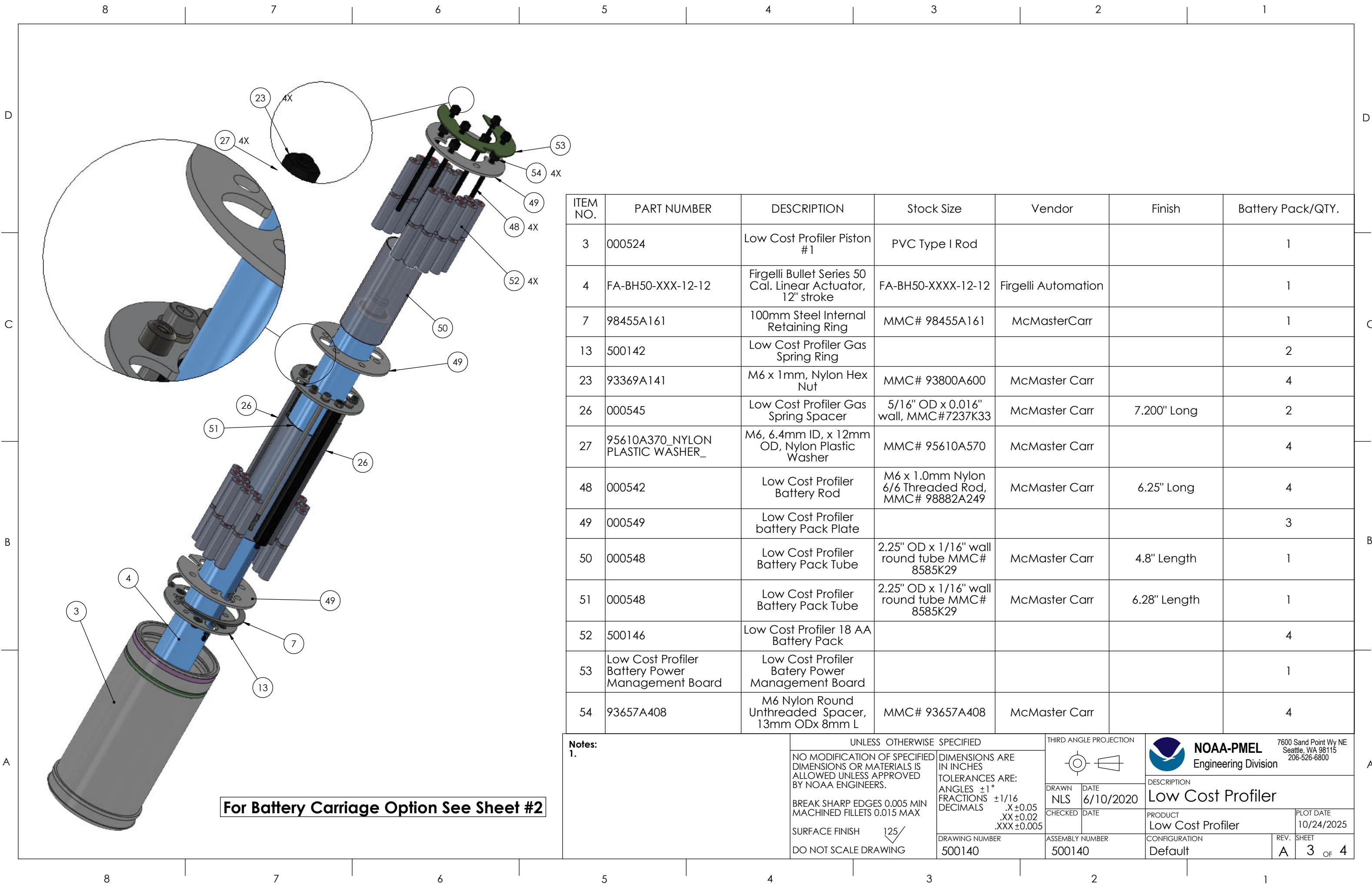
PRODUCT Low Cost Profiler

CONFIGURATION Default

REV. A

SHEET 2 OF 4

PLOT DATE 10/24/2025



ITEM NO.	PART NUMBER	DESCRIPTION	Stock Size	Vendor	Finish	Battery Pack/QTY.
3	000524	Low Cost Profiler Piston #1	PVC Type I Rod			1
4	FA-BH50-XXX-12-12	Firgelli Bullet Series 50 Cal. Linear Actuator, 12" stroke	FA-BH50-XXXX-12-12	Firgelli Automation		1
7	98455A161	100mm Steel Internal Retaining Ring	MMC# 98455A161	McMasterCarr		1
13	500142	Low Cost Profiler Gas Spring Ring				2
23	93369A141	M6 x 1mm, Nylon Hex Nut	MMC# 93800A600	McMaster Carr		4
26	000545	Low Cost Profiler Gas Spring Spacer	5/16" OD x 0.016" wall, MMC#7237K33	McMaster Carr	7.200" Long	2
27	95610A370_NYLON PLASTIC WASHER_	M6, 6.4mm ID, x 12mm OD, Nylon Plastic Washer	MMC# 95610A570	McMaster Carr		4
48	000542	Low Cost Profiler Battery Rod	M6 x 1.0mm Nylon 6/6 Threaded Rod, MMC# 98882A249	McMaster Carr	6.25" Long	4
49	000549	Low Cost Profiler battery Pack Plate				3
50	000548	Low Cost Profiler Battery Pack Tube	2.25" OD x 1/16" wall round tube MMC# 8585K29	McMaster Carr	4.8" Length	1
51	000548	Low Cost Profiler Battery Pack Tube	2.25" OD x 1/16" wall round tube MMC# 8585K29	McMaster Carr	6.28" Length	1
52	500146	Low Cost Profiler 18 AA Battery Pack				4
53	Low Cost Profiler Battery Power Management Board	Low Cost Profiler Batery Power Management Board				1
54	93657A408	M6 Nylon Round Unthreaded Spacer, 13mm ODx 8mm L	MMC# 93657A408	McMaster Carr		4

Notes:
1.

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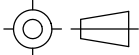
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DIMENSIONS ARE IN INCHES
TOLERANCES ARE:
ANGLES ±1°
FRACTIONS ±1/16
DECIMALS .X±0.05
.XX±0.02
.XXX±0.005

DRAWING NUMBER
500140

THIRD ANGLE PROJECTION



DRAWN NLS
DATE 6/10/2020
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DATE

ASSEMBLY NUMBER
500140



NOAA-PMEL
Engineering Division

7600 Sand Point Wy NE
Seattle, WA 98115
206-526-6800

DESCRIPTION

Low Cost Profiler

PRODUCT

Low Cost Profiler

CONFIGURATION

Default

PLOT DATE
10/24/2025

REV. SHEET

A 3 OF 4

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Inner Assembly Installation Instructions:

1. Unscrew the Actuator #4's rod end fitting from the rod, being careful to not unscrew the rod from the actuator. After rod end is removed, double check that the rod still fully screwed into the actuator.

2. Lightly Coat # 36 with vibratight threadlocker, wait 15+min & screw into #4. Double check that #4's rod is fully screwed into the actuator.

3. Lightly grease with Aqua Shield & install #46 into #3 using Rod Seal Installation Tool: MMC#45015K23

4. Lightly grease with Aqua Shield & install #45 onto #3 using Piston1 Seal Installation Tool A Assembly, 500195, & Tool B, 000578.

5. Clean #25 threads with IPA and coat with Vibratight threadlocker, wait 15+ min & attach #4 to #5 with #25.

6. Install #42 into #3.

7. <PARA >Generously grease interior bore of #3 and OD of #5 with Aqua Shield grease.

8. Install Piston2 Seal Installation Tool, 000574, onto #5.

9. Install #44 onto #5 so that the ends of #44 join in the middle of #5's wire clearance cutout.

10. Insert #5 into #3, you can use a pair of needle nose pliers to gently pull on the end of #44 during installation. After installation, move actuator cable to the inside of the wear ring.

11. Remove Piston2 Seal Installation Tool, 000574.

12. Pass the #4 actuator's cable thru lower #13, using of of the gas spring mounting holes which are 90° from the ear cutouts.

13. Make sure piston #5 is fully extended in #3, Fully seat #13 in #3, pull the cable tight and mark where it exits the top of #13. Remove #13, then measure 1" from mark on cable towards the cable's bitter end and install a ziptie around the cable (MMC#7130K13), tight enough that it does not move on the cable.

14. Retract piston #5 and install lower #13 as shown into #3. Mark location of #13 gas spring ear cutouts on #3's ID so that they align with the 2 counterbored slots in the top of #5. Then fully extend piston #5. Secure #13 to #3 with #7, making sure to install #7 so that its free ends are 180° from the actuator cable.

15. Add a second ziptie (MMC#7130K13) around the cable ~ 0.25" above where cable exits #13 plate.

16. 2X: Bolt #6 to Upper #13 as shown with #17.

17. 4x: Coat #15 threads with Vibratight threadlocker, wait 15+ min, install #16 & pass thru upper #13.

18. 2X: Pass #15 thru #26, place on opposite sides from each other.

19. Lower Battery Installation: **IMPORTANT: Make sure that #6 gas spring free ends are aligned with the 2 counterbored slots in #5.**

1. Battery Carriage: Slide lower 4 battery packs #14 over #13-15-16-26 assembly & bolt (hand tight) to #3-13-7-4-5 assembly.

2. Battery Pack: Slide #49 & #51 over the #3-13-7-4 assembly. Then install 2X lower battery packs #52, 500146, use foam to sandwich battery packs & bolt #13-15-16-26 assembly to #3-13-7-4 assembly. Route actuator cable thru slot in #49.

20. Tighten bolts, slowly working your way between opposing bolt side (like tightening a flange) until #26s are hard sandwiched between upper and lower #13s. Double check that the dimension between the bottom of the upper #13 and the top face of #3 is 6.42" (as shown at both gas spring mounting locations). Adjust to meet the 6.42" dimension if needed.

21. Upper Battery Installation:

1. Battery Carriage: 4X: Bolt upper battery packs #14 & power management board #53 to upper #13 plate with #48 & #23& #27, use Loctite 425 threadlocker on all connections. Route actuator cable along slots in the OD of lower #14, thru the ear cutout in upper #13, and along slots in upper #14.

2. Battery Pack: Slide second #49 & #50 over #4. Then install 2X upper battery packs #52, 500146, use foam to sandwich battery packs. Slide third #49 over #4. Install 4X #54 & Battery management board #53. Bolt #49-50-52-53-53 assembly to upper #13 plate with 4X: #48 & #23& #27, use Loctite 425 threadlocker on all connections. Route actuator cable along gas spring, thru the ear cutout in upper #13 and 2X upper #49s.

22. Remove any slack from, and securely tape, the actuator's cable tight around the blue actuator housing ~1" down from where the rod exits the blue housing.

Sensor Endcap Installation Instructions:

1. Insert #32 into #30 and secure with #33.

2. 3X: Assemble sensors into 000531 sensor ports.

3. Place tape over top of 000531 pressure port hole, #8, place face down and fill sensor cup with API/GL-4 Multi-Purpose Hypoid Gear oil. Gently install #24 into #8 using the oil as a lubricant, secure in place with #18. Remove tape.

4. Install #40 onto #2, #39 onto #30, and 4X #38 onto 00531, lightly grease O-rings and the IDs of the #2 ports with Dow Corning #4 silicone grease. Install into #2 as shown and secure with 4X #19 & #34.

5. Attach #37 to #2 with 3X: #35 & #31, use Loctite 425 on all connections.

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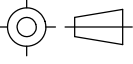
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DRAWING NUMBER
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THIRD ANGLE PROJECTION



DRAWN NLS
CHECKED DATE

DATE
6/10/2020

ASSEMBLY NUMBER
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NOAA-PMEL
Engineering Division

7600 Sand Point Wy NE
Seattle, WA 98115
206-526-6800

DESCRIPTION
Low Cost Profiler

PRODUCT
Low Cost Profiler

CONFIGURATION
Default

REV. A

PLOT DATE
10/24/2025

SHEET
4 OF 4

8

7

6

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For Battery Carriage Option See Sheet #2

For Battery Pack Option See Sheet #3